

To: The Director, Outreach & Academic Collaboration Cell, ISRO Headquarters, Bengaluru
From: Akshat Verma · akshatverma26062004@gmail.com · +91 98912 58917

Summary

Since 2020, people across cultures report that time feels faster — that 2020–2025 passed as "a blur." The physical clock did not change; subjective time did. This proposal frames that shift as a **measurable** phenomenon arising from three interacting, well-studied factors: reduced novelty and memory density, disrupted circadian rhythms, and continuous digital stimulation. It outlines a study — suited to collaboration through ISRO's Outreach & Academic Collaboration Cell — combining psychometric testing, satellite-derived environmental data, and computational modelling to quantify a "Perceived Time-Compression Index," with direct relevance to astronaut psychology in long-duration and isolated missions.

Background & Rationale

Memory density and retrospective time. The brain estimates how long a period felt partly from how many distinct memories it holds. Confinement and routine reduced novelty, lowering memory density; sparse memory makes a stretch of time feel shorter in hindsight — the documented "holiday paradox" (Eagleman 2008; Wittmann 2016).

Circadian disruption. Lockdowns altered exposure to natural light — the primary signal entraining the suprachiasmatic nucleus, the body's master clock. Irregular light–dark cycles degrade circadian regularity, which is linked to both time estimation and mood (Roenneberg 2012).

Digital stimulation and attention. Sharply increased screen time and rapid feedback loops fragment attention and raise tolerance thresholds for delay, plausibly compressing the felt pace of time. These are measurable behavioural variables — deliberately framed without cultural commentary about "resilience" or generational decline, which is opinion rather than measurement.

Hypothesis

> **H.** Post-2020 subjective time acceleration is driven by the interaction of reduced novelty/memory density, circadian disruption, and digital overstimulation — each independently measurable, and together expressible as a single quantifiable index.

Proposed Methodology

1. Psychometric measurement. Administer validated time-estimation, retrospective-duration, memory-density and attention-span instruments to post-2020 cohorts, stratified by age and digital exposure.

2. Environmental correlation (ISRO's contribution). Use satellite-derived nighttime-light and light-pollution data (e.g., VIIRS-class products) as a proxy for changes in the circadian light environment across regions and time — a

concrete, defensible use of Earth-observation assets. This replaces any claim to measure "consciousness" or electromagnetic "resonance," which lies outside scientific support.

3. Computational model. Integrate cognitive-load, circadian, and digital-exposure variables into a Perceived Time-Compression Index (PTCI), validated against the psychometric data.

4. Collaboration. ISRO Earth-Observation and atmospheric divisions provide environmental data; partner neuroscience and psychology institutions provide testing and analysis.

Relevance to ISRO

Isolation, confinement and altered light cycles in long-duration and analog missions reproduce many of the same conditions studied here. Understanding time-perception distortion on Earth directly supports astronaut psychological preparation and scheduling for India's human-spaceflight programme, and informs habitats and routines that protect circadian and cognitive health.

Philosophical & Cultural Framing

As cultural framing — not a measured mechanism — Indian thought has long modelled time as cyclical rather than linear, each *yuga* carrying a distinct tempo of life and awareness. The intuition that an age of abundant information but shallow assimilation should *feel* faster resonates with that tradition. It is offered here as a source of hypotheses and conceptual richness — in the spirit of bringing Indian knowledge systems into dialogue with empirical inquiry — and explicitly not as a variable to be measured by instruments.

A Note on Method

This proposal keeps two layers distinct. The **testable** layer — memory, circadian rhythm, attention, and their satellite-observable environmental correlates — uses standard psychological and Earth-observation methods. The **philosophical** layer — cyclical-time cosmology — is cultural framing only. The study makes no claim to measure "global consciousness," Schumann-resonance entrainment, or electromagnetic effects on the mind; those ideas lack scientific support and are deliberately excluded from the methodology.

References

Eagleman, D. M. (2008). Human time perception and its illusions. *Current Opinion in Neurobiology*, 18(2), 131–136. · Wittmann, M. (2016). *Felt Time: The Psychology of How We Perceive Time*. MIT Press. · Roenneberg, T. (2012). *Internal Time: Chronotypes, Social Jet Lag, and Why You're So Tired*. Harvard University Press. · Postman, N. (1985). *Amusing Ourselves to Death*. Penguin. · Bhagavata Purana, Book 12 — yuga cycles (cited under cultural framing).